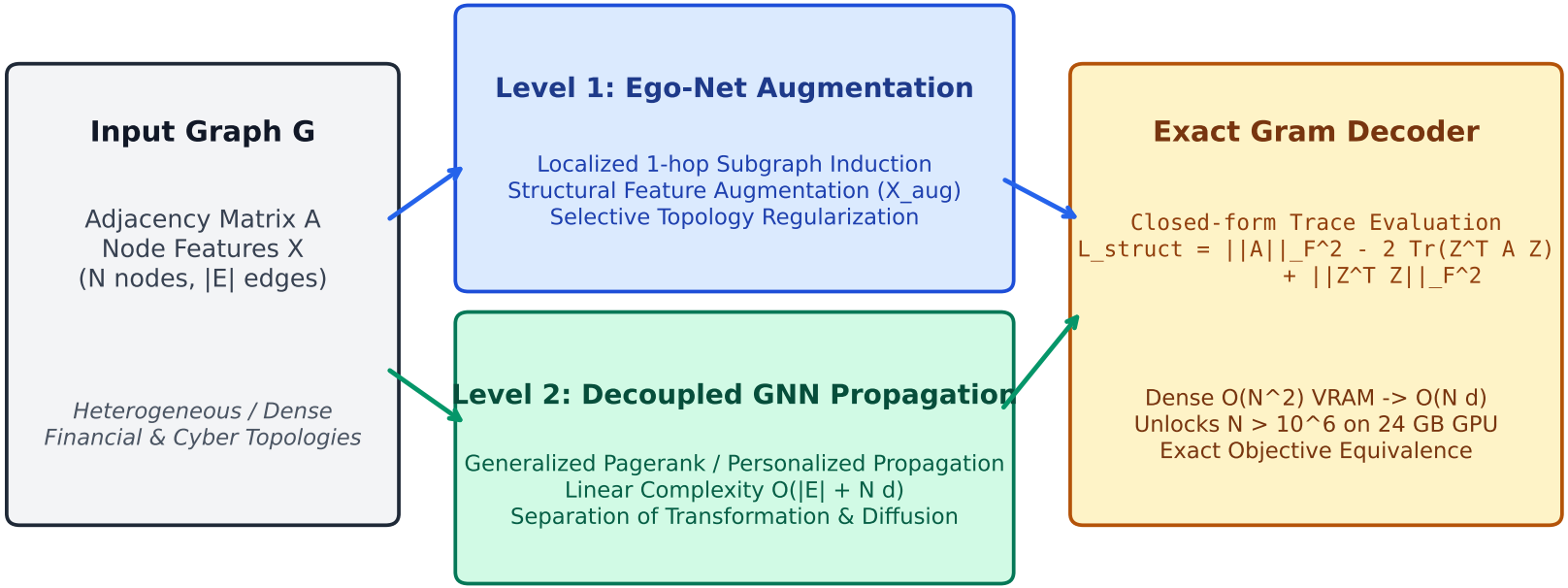
Input Graph G
Adjacency Matrix A
Node Features X
(N nodes, |E| edges)
Heterogeneous / Dense
Financial & Cyber Topologies
Level 1: Ego-Net Augmentation
Localized 1-hop Subgraph Induction
Structural Feature Augmentation (X_aug)
Selective Topology Regularization
Level 2: Decoupled GNN Propagation
Generalized Pagerank / Personalized Propagation
Linear Complexity O(|E| + N d)
Separation of Transformation & Diffusion
Exact Gram Decoder
Closed-form Trace Evaluation
L_struct = ||A||_F^2 - 2 Tr(Z^T A Z)
+ ||Z^T Z||_F^2
Dense O(N^2) VRAM -> O(N d)
Unlocks N > 10^6 on 24 GB GPU
Exact Objective Equivalence